VF-G03 Whole Shoe Wash

Test Method for Whole Shoe Wash	Test method # VF- G03 Page 1 of 3
Date Issued: August 12, 2024 **Revision #: 3**	**Reason for Revisions:** **Add sandal/slippers into test scope**

PURPOSE AND SCOPE: To assess the color and dimensional stability of footwear during washing but not the ease of soil removal.

REFERENCES: BS EN 26330 / SATRA TM 158 EQUIPMENT

AND MATERIAL:

1. Automatic Washer and Dryer
2. Laboratory Grade Thermometer (0° to 100°C), with an accuracy of +/-1 ^{0}C
3. Graduated glass cylinder, capable of measuring 50+/-1 cm^3
4. AATCC Chromatic transference scale
5. Detergent as specified in BS EN 26330
6. Stick length gauge, toe spring gauge or steel ruler
7. Assembled shoe, sandal, slippers.

SAMPLE PREPARATION:

1. Condition the sample for at least 48 hours at standard laboratory environment at 23.0+/-2.0 ˚C temperature and 55 ±10 % Relative Humidity to allow the cement to crosslink completely.
2. All hang tags, tissue paper, and removable foot-bed should be removed from the test samples.
3. Lace and tie the shoe in a normal manner (Typical lace opening should be 3 cm).
4. Shoes with metal attachments (buckles, studs) should be placed in a sealed cloth bag to minimize the risk of damage to the shoes and interior of washing machine.

TEST PROCEDURE:

1. Place the shoe (s) inside washing machine. Only one type of shoe may be washed at a time, different types and colors of shoes should be washed separately.
2. The washer shall be operated as follows:
    - Water level setting-HIGH
    - Machine control setting- Footwear (optional – depends on the machine type)
    - Water temperature setting @ RT or maximum at 30 ^{0}C (optional – depends on the machine type)

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3. Set water temperature @ 30ºC or at RT if the washing machine doesn't have provision on temperature control.
4. Add detergent, at approx. 4 grams for every ½ pair of shoes.
5. The washing cycle: Wash, drain, wash, drain, dry.
6. Dry at temperature between 50 - 60ºC for 30 minutes. If the machine doesn't have provision for drying, let it dry at room temperature. Sandal/slippers after washing let them dry at room temperature.
7. When fully dried, examined for visual changes in conditions and these changes noted.

EXAMINATION OF SHOES:

Shoes shall be examined and noticeable changes in visual appearance, from the shoe before testing recorded.

1. Appearance of upper materials, i.e., abrasion, fraying, finish peeling, surface cracking, color changes/migration, bleeding, staining, etc. Slight abrasion is acceptable, but should in no case result in wear through at any point on the upper.
2. Color migration, staining or color bleeding from one component to another is totally unacceptable.
3. Shoes must be checked for signs of outsole, midsole or upper delamination or separation.
4. No stitching failures (broken threads, obvious wear, loose lining, seam failures, etc.) Nylon bonded 6.6 thread is required throughout all shoes (exception as specified)
5. For sock-liners cemented in place the sock-liner shall be in place throughout the entire wash, drain and dry cycles
6. Shrinkage of upper materials, at the end of the test cycles, shall be minimal to avoid serious deformation of the upper.
7. Silk screening should be done with paints that are colorfast, and should not result in bleeding, etc. ***There should be no sign of delamination at the end of the test cycle.*** **No need to evaluate delamination or peel off which cause by material shrinkage.**
8. There should be no unraveling of any material/component at the end of the test cycle
9. Check shoe laces for color bleeding, color changes, fraying or unraveling, etc.
10. All PU materials, or PU coated materials should be examined for any visible signs of hydrolysis occurring.

NOTE: Total testing time for the complete cycle at 80±5 minutes.

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REPORT:

1. Evaluate appearance of footwear after the end of testing cycle.
2. Any major failure noted during the test cycle, should immediately terminate the test and appropriate actions should be taken, this is to assure that substandard product will not reach our customer, and that production is stopped until corrective measure can be taken and verified.

Note:

1. ***After the above examination is completed, the footwear or shoe MAY be cut in half (in the length direction) and the various bottom bonds assessed, lasting board, lasting allowance, heel counter, outsole to midsole, midsole to upper, etc. There should be no loss in the strength of these bonds***. **2.** ***May conduct bonding test strength if required.***

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